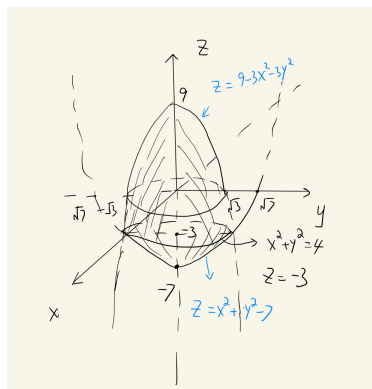
MAST20009 Vector Calculus

Practice Class 7 Solutions

1. (a) The two paraboloids meet when $x^2 + y^2 - 7 = 9 - 3x^2 - 3y^2$.
Then $4x^2 + 4y^2 = 16 \implies x^2 + y^2 = 4$ and $z = -3$.

(b)



(c) Bounding z first gives $x^2 + y^2 - 7 \le z \le 9 - 3x^2 - 3y^2$.
Projecting D onto the xy-plane gives a disk $x^2 + y^2 \le 4$.
Using vertical strips we have

$$D = \{(x, y, z) \in \mathbb{R}^3 \mid -2 \le x \le 2,\ -\sqrt{4 - x^2} \le y \le \sqrt{4 - x^2},\ x^2+y^2-7 \le z \le 9-3x^2-3y^2\}.$$

So,

$$\begin{aligned}
\iiint_D x\,dV &= \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{x^2+y^2-7}^{9-3x^2-3y^2} x\,dz\,dy\,dx \\
&= \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \left(16x - 4x^3 - 4xy^2\right) dy\,dx \\
&= \int_{-2}^{2} \left[16xy - 4x^3y - \frac{4xy^3}{3}\right]_{y=-\sqrt{4-x^2}}^{y=\sqrt{4-x^2}} dx \\
&= \int_{-2}^{2} \left[32x(4-x^2)^{1/2} - 8x^3(4-x^2)^{1/2} - \frac{8x}{3}(4-x^2)^{3/2}\right] dx \\
&= \int_{-2}^{2} (4-x^2)^{1/2} \left(32x - 8x^3 - \frac{32x}{3} + \frac{8x^3}{3}\right) dx \\
&= \int_{-2}^{2} (4-x^2)^{1/2} \left(\frac{64x}{3} - \frac{16x^3}{3}\right) dx \\
&= \int_{-2}^{2} \frac{16x}{3}(4-x^2)^{3/2} dx = \left[-\frac{16}{15}(4-x^2)^{5/2}\right]_{x=-2}^{x=2} = 0.
\end{aligned}$$

2. Using polar coordinates we can describe R by

$$0 \le r \le 2, \quad 0 \le \theta \le \pi.$$

Since $x^2 + y^2 = r^2$, then

$$\iint_R \left(x^2+y^2\right)^{\frac{3}{2}} dx\,dy \;=\; \int_0^{\pi}\int_0^{2} r^3 r dr\, d\theta = \int_0^{\pi}\left[\frac{r^5}{5}\right]_{r=0}^{r=2} d\theta = \frac{32}{5}\int_0^{\pi} d\theta = \frac{32\pi}{5}.$$

3. The image of the parallelogram D under the change of variables is a square D^* in the uv-plane with vertices at $(0,0), (0,1), (1,0), (1,1)$.

Therefore, D^* can be described by $0 \le u \le 1$ and $0 \le v \le 1$.

Since $x = 2u + v$ and $y = 2u + 2v$,

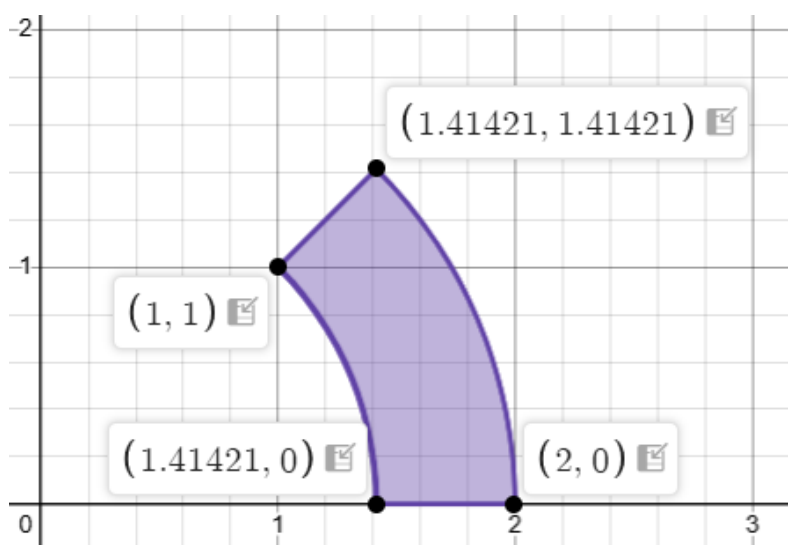
$$\text{Jacobian} \;=\; \det\begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix} \;=\; 2.$$

$$\iint_D \left(x^2+y^2\right) dx\,dy \;=\; \iint_{D^*} \left((2u+v)^2 + (2u+2v)^2\right) 2du\,dv$$

$$= \int_0^1\int_0^1 \left(16u^2 + 24uv + 10v^2\right) du\,dv = \int_0^1 \left[\frac{16u^3}{3} + 12u^2v + 10uv^2\right]_{u=0}^{u=1} dv$$

$$= \int_0^1 \left(\frac{16}{3} + 12v + 10v^2\right) dv = \left[\frac{16v}{3} + 6v^2 + \frac{10v^3}{3}\right]_{v=0}^{v=1} = \frac{44}{3}.$$

4. (a)



(b) We use the standard change of variables for polar coordinates:

$$x = r\cos\theta \qquad \text{and} \qquad y = r\sin\theta.$$

In terms of polar coordinates, the region $\mathcal{L}$ is bounded by circles of radius $r = \sqrt{2}$ and $r = 2$, and by rays of angle $\theta = 0$ and $\theta = \pi/4$. The Jacobian is r. Our expression then becomes

$$M = \iint_{\mathcal{L}} \rho(x,y)\ \mathrm{d}A = \int_0^{\frac{\pi}{4}}\int_{\sqrt{2}}^{2} \rho(r\cos\theta, r\sin\theta)\ r\ \mathrm{d}r\mathrm{d}\theta.$$

(c) When

$$\rho(x,y) = \frac{2xy}{x^2+y^2},$$

we substitute in $x = r\cos\theta$ and $y = r\sin\theta$ to obtain

$$\rho(r\cos\theta, r\sin\theta) = \frac{2r^2\sin\theta\cos\theta}{r^2} = 2\sin\theta\cos\theta = \sin(2\theta).$$

Then our integral is

$$\begin{aligned}
M &= \int_0^{\frac{\pi}{4}} \int_{\sqrt{2}}^{2} \rho(r\cos\theta, r\sin\theta)\ r\ \mathrm{d}r\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{4}} \int_{\sqrt{2}}^{2} r\sin(2\theta)\ \mathrm{d}r\mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{4}} \left[\frac{1}{2}r^2\sin(2\theta)\right]_{r=\sqrt{2}}^{r=2} \mathrm{d}\theta \\
&= \int_0^{\frac{\pi}{4}} \sin(2\theta)\ \mathrm{d}\theta \\
&= \left[-\frac{1}{2}\cos(2\theta)\right]_{\theta=0}^{\theta=\frac{\pi}{4}} \\
&= -\frac{1}{2}(0-1) \\
&= \frac{1}{2}.
\end{aligned}$$

5. (a)

(b) Since y is bounded above by two different curves, we must divide the region in two at the point where the curves meet. This point is $(\sqrt{2}, \sqrt{2})$, so divide the region along the line $x = \sqrt{2}$.

In the new left region, we have $0 \le y \le x$ and $0 \le x \le \sqrt{2}$. In the new right region, we have $0 \le y \le \sqrt{4-x^2}$ and $\sqrt{2} \le x \le 2$. So the integral becomes

$$J = \int_0^{\sqrt{2}} \int_0^{x} \frac{y}{x} e^{x^2+y^2}\ \mathrm{d}y\mathrm{d}x\ +\ \int_{\sqrt{2}}^{2} \int_0^{\sqrt{4-x^2}} \frac{y}{x} e^{x^2+y^2}\ \mathrm{d}y\mathrm{d}x.$$

(c) Our region can be described using polar coordinates with bounds $0 \leq r \leq 2$ and $0 \leq \theta \leq \pi/4$. Rewriting the integral J appropriately (don't forget the Jacobian!) and then evaluating gives

$$\begin{aligned} J &= \int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} \frac{y}{x} e^{x^2+y^2} \, \mathrm{d}x\mathrm{d}y \\ &= \int_0^2 \int_0^{\frac{\pi}{4}} \frac{r\sin\theta}{r\cos\theta} e^{r^2} \, r \, \mathrm{d}\theta\mathrm{d}r \\ &= \int_0^2 \int_0^{\frac{\pi}{4}} re^{r^2} \frac{\sin\theta}{\cos\theta} \, \mathrm{d}\theta\mathrm{d}r. \end{aligned}$$

Use substition: let $u = \cos\theta$, then $\frac{du}{d\theta} = -\sin\theta$. When $\theta = 0$ we have $u = 1$ and when $\theta = \pi/4$ we have $u = 1/\sqrt{2}$. We then rewrite and continue to evaluate J as follows:

$$\begin{aligned} J &= \int_0^2 re^{r^2} \int_1^{\frac{1}{\sqrt{2}}} \frac{\sin\theta}{u} \frac{-1}{\sin\theta} \, \mathrm{d}u\mathrm{d}r \\ &= \int_0^2 re^{r^2} \int_1^{\frac{1}{\sqrt{2}}} -\frac{1}{u} \, \mathrm{d}u\mathrm{d}r \\ &= \int_0^2 re^{r^2} [-ln|u|]_{u=1}^{u=\frac{1}{\sqrt{2}}} \, \mathrm{d}r \\ &= \int_0^2 re^{r^2} \left(\frac{1}{2}ln(2) + 0\right) \, \mathrm{d}r \\ &= \frac{1}{2}ln(2) \int_0^2 re^{r^2} \, \mathrm{d}r. \end{aligned}$$

This time using the substitution $v = r^2$ with $\frac{dv}{dr} = 2r$ and corresponding bounds $0 \leq v \leq 4$, we get

$$\begin{aligned} J &= \frac{1}{2}ln(2) \int_0^4 re^v \frac{1}{2r} \, \mathrm{d}v \\ &= \frac{1}{4}ln(2) \int_0^4 e^v \, \mathrm{d}v \\ &= \frac{1}{4}ln(2) \, [e^v]_{v=0}^{v=4} \\ &= \frac{1}{4}ln(2)[e^4 - 1]. \end{aligned}$$